Wildfire Detection and Mapping by Satellite With an Enhanced Configuration of the Normalized Hotspot Indices: Results From Sentinel-2 and Landsat 8/9 Data Integration

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***Abstract*— The Operational Land Imager (OLI) and the Multispectral Instrument (MSI), respectively, aboard Landsat-8/9 (L8/9) and Sentinel-2 (S2) satellites, by providing data in near infrared (NIR) and short-wave infrared (SWIR) bands, with a mid-high spatial resolution (20/30 m), enable the identification, mapping, and characterization of high-temperature features. Here, we exploit this potential by presenting and testing the normalized hotspot indices algorithm tailored to fire mapping (NHI-F). Results were achieved by investigating the devastating fire events occurring in California and Hawaii islands (USA), Yellowknife (Canada), Tenerife islands (Spain), North Attica (Greece), and Northern Territory (Australia), during the intense fire seasons of 2023, show the high performance of the NHI-F in detecting and mapping wildfires, despite multispectral misregistration and striping effects affecting S2-MSI imagery. These effects may be directly minimized from the used indices, as demonstrated in this work. By investigating the wildfires of Yellowknife and California by means of L8/9 OLI/OLI2 data, we found that the NHI-F flagged up to 99% of fire pixels detected by the operational Landsat Fire and Thermal Anomaly (LFTA) product. Moreover, the additional fire pixels from NHI-F (up to 70% in night-time conditions) better detailed the fire fronts and provided unique information also about some small-fire outbreaks. The analysis of fire dynamics, performed integrating L8/9 (nighttime/daytime) and S2 (daytime) observations, demonstrates that the NHI-F configuration may highly support the fire monitoring activities at different spatial scales, complementing information from systems using satellite data at high-temporal/low spatial resolution.**

***Index Terms*— Fires, Landsat 8/9 (L8/9), normalized hotspot indices algorithm tailored to fire mapping (NHI-F), Sentinel-2 (S2).**

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I. Introduction

In THE last decade, fires have destroyed millions of hectares of forest across the world affecting both the environment and climate (e.g., [1], [2]). Fires may lead to significant changes in vegetation, threaten biodiversity, and determine the emission of large amounts of carbon dioxide and other climate-altering compounds into the atmosphere, such as methane, nitrogen oxides, and black carbon (e.g., [3], [4]). The global emissions from wildfires were estimated at 1.8 Gt C yr^{-1} during the period 2003–2022 [5], while during the March 2023–February 2024 fire season, when 3.9×10^6 km^2 burned globally, the increment in the carbon emission (16% above average) was affected by the record emissions in the Canadian forest [6]. Indeed, wildfires have shown a significant increment in the high-latitude regions, where spruce, fir, larch, pine, birch, and aspen dominate the boreal forest [7], owing to unusually warm temperatures and drought conditions [8]. These factors strongly affect the fire behavior, frequency, and fire season length (e.g., [9], [10], [11]). Globally, the fire weather season length has increased by 14 days $year^{-1}$ over the period 1979–2019, while the annual days exceeding the 95th percentile fire weather index value have increased by 10 days $year^{-1}$ [6]. In the next years, the occurrence, frequency, and severity of wildfires are expected to further increase because of global warming (e.g., [12], [13]).

Fire detection systems capable to support fire management activities in an efficient and timely way are then crucial to mitigate the impact of wildfires on the environment and climate. Satellite remote sensing played, for decades, a key role in this context, thanks to multispectral data provided at different spatial, spectral, and temporal resolutions (e.g., [14], [15], [16]).

The Advanced Very High-Resolution Radiometer (AVHRR) and Moderate Resolution Imaging Spectroradiometer (MODIS) sensors, by providing medium infrared (MIR, 3–5 μm) and thermal infrared (TIR, 8–12 μm) data, with a good compromise between spatial and temporal resolution (e.g., 1 km/12 h), were widely used for fire detection purposes (e.g., [17], [18], [19], [20]). MODIS allows for estimates of the fire radiative power (FRP), also through a MIR channel having a higher saturation temperature

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(e.g., [21], [22], [23], [24], [25], [26]). With a similar spatial resolution, the Visible and Infra-Red Radiometer (VIRR), onboard the FengYun-3C satellite, has shown its potential in active fire detection (AFD) as well [27]. Other recent polar orbiting sensors such as Visible Infrared Imaging Radiometer Suite (VIIRS) and Sea and Land Surface Temperature Radiometer (SLSTR), thanks to the improved spectral and spatial resolution (e.g., up to 375 m in the infrared bands for VIIRS), have further extended the detection capabilities of wildfires (e.g., [26], [28], [29]). Again, many authors exploited the 15 min temporal resolution of Spinning Enhanced Visible and Infrared Imager (SEVIRI), aboard Meteosat Second Generation (MSG) geostationary satellites, to monitor wildfires in near-real time (NRT) (e.g., [30], [31]). This sensor, such as Advanced Himawari Imager (AHI) and Advanced Baseline Imager (ABI), respectively, aboard Himawari-8 and Geostationary Operational Environmental Satellites-R Series (GOES-R) satellites, was used also to analyze the fire diurnal cycles and estimate the fire radiative energy (FRE) (e.g., [32], [33], [34]), which can be used to retrieve information about the atmospheric fire emission (e.g., [35], [36]).

Sensors such as Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER), thanks to short-wave infrared (SWIR) bands at 30 m spatial resolution, enabled a more accurate analysis of the active fire areas [37]. Nevertheless, since 2008 ASTER SWIR detectors are no longer functioning; the Operational Land Imager (OLI) and Multispectral Instrument (MSI), respectively, aboard Landsat-8 and Sentinel-2 (S2) satellites, currently guarantee the continuity of SWIR observations (e.g., [26], [38]).

In this work, we exploit near infrared (NIR) and SWIR data from those sensors to test an enhanced configuration of the normalized hotspot indices (NHI) algorithm [39] tailored to active fire mapping. We analyze, for this purpose, some devastating fire events occurring across the world during the intense fire season of 2023; the hottest calendar year in global temperature data records since 1850 [40].

Results of the comparison with the operational Landsat Fire and Thermal Anomaly (LFTA) product [41], and the integration with information from satellite-based systems performing the NRT monitoring of active fires, are shown and discussed.

A confutation analysis performed over the Brady Hot Spring (Nevada, USA) geothermal area devoted to further assessing factors affecting the NHI algorithm tailored to fire mapping (NHI-F) performance is then reported in the Appendix.

II. Background

Numerous satellite fire-detection methods were developed over the years to detect smoldering and flaming fires, showing maximum temperatures around 500 and 1200 K, respectively [42]. Starting from the first single-channel and multichannel algorithms, analyzing the brightness temperatures measured in the MIR and/or TIR bands of AVHRR and MODIS, generally through fixed threshold tests used globally, more complex detection schemes were developed to improve fire detection from space (e.g., [18]). Contextual methods compare the MIR brightness temperature of the potential fire pixels with the background (e.g., in terms of spatial average and standard deviation) to confirm/reject fires. Moreover, they generally use some additional spatial/spectral filters (e.g., cloud and land-sea mask) to minimize false detections (e.g., [24]). Among those methods, the MODIS fire detection algorithm showed high performance in different observational conditions [24], [43]. This method is used operationally by the Fire Information for Resource Management System (FIRMS) on a global scale (https://firms.modaps.eosdis.nasa.gov) and represents the basis of active fires for European Forest Fire Information System (EFFIS) (https://forest-fire.emergency.copernicus.eu/).

Multitemporal techniques by analyzing satellite data in both space and time showed further improvements in wildfire detection (e.g., [19], [23], [27]). Among those methods, Robust Satellite Techniques for FIRES detection and monitoring (RST-FIRES) was tested with success also in the framework of early warning contexts, demonstrating the capacity of SEVIRI to support the fire management activities at the regional/local scale, despite the low spatial resolution (3 km at nadir) (e.g., [33]). Other authors developed some hybrid approaches (e.g., integrating the Kalman filter) to detect fires before estimating the FRP (e.g., [44]). More accurate information about wildfires, in terms of temperature and size, was then retrieved from ASTER data [37], which were used also to validate the MODIS and GOES fire products (e.g., [45], [46]).

The OLI instrument, thanks to the improved spectral and spatial features than Thematic Mapper (TM) and Enhanced TM (ETM+) of the prior Landsat series, encouraged the development of novel algorithms to map active fires (e.g., [38], [47], [48], [49], [50]). The method proposed in [38] analyzes the top of the atmosphere (TOA) reflectance, measured in the visible (VIS), NIR, and SWIR bands of OLI to detect wildfires in the daytime, and integrates a multitemporal approach to discriminate fires from persistent heat sources. A simple threshold test on SWIR radiance is then used in night-time conditions [38]. Among the methods of analyzing S2-MSI imagery, the active fire detection (AFD-S2) algorithm [52] and thermal anomaly index (TAI) [53] showed good potential in the identification of active fire areas, with some limitations (e.g., reduced sensitivity under certain fire conditions [53]). Other spectral indices developed for the fire mapping were tested with good results in specific fire conditions (e.g., [54], [55]).

Recently, several studies have shown the potential of machine learning and deep learning models in better detecting small fires in comparison with traditional fire detection techniques (e.g., [56], [57]). Examples are a custom-optimized convolutional neural network (CNN) model using Landsat data [58], and the implementation of a big wildfire dataset of S2 images to train various deep learning models for fire monitoring [59]. Despite this capacity, the above-mentioned models are not always suited for operational scenarios and large spatial scales also because of the high computing power required (e.g., [60], [61]).

The method presented and tested in this work offers the same advantages as the NHI algorithm, which is used operationally for the worldwide mapping of thermal volcanic features and gas flaring sources (e.g., [39], [62], [63], [64]). They include the use of a simple single-image detection

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TABLE I
MAIN FEATURES OF THE WILDFIRES INVESTIGATED IN THIS WORK

Fire Location	*Climate*	*Vegetation cover*	*Tot Burned area (ha)*	*Satellite scenes*
North Attica (Greece)	Mediterranean	Forest	6193	23 August 2023
Tenerife (Spain)	Mediterranean	Forest	15,000	18 August 2023
Hawaii (USA)	Tropical	Tall Grass	438	8 August 2023
California (USA)	Temperate	Forest	32,900	15–30 August 2023
Darwin Region (Australia)	Tropical	Savannah	440	13 July 2023
Yellowknife (Canada)	Subarctic	Boreal Forest	50,750	9 August 2023

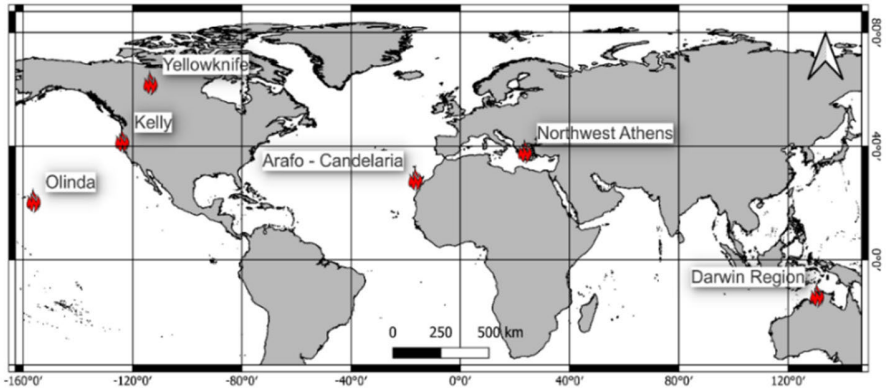


Fig. 1. Geographic location of the active fires investigated in this work.

scheme (see Section V), which can be easily implemented also in geospatial platforms such as Google Earth Engine (GEE) [65].

III. TEST CASES

During 2023, many devastating fires occurred in the USA, Canada, Hawaii, and Europe due to the extrem weather conditions affecting a large part of the Northern hemisphere (e.g., [66]). In terms of global emissions, 1.9–2.3 Gt C yr^{-1} were estimated during January–October 2023. These values were 10%–28% above the 2003–2022 average [5].

The largest wildfire recorded in the United States occurred in California, covering about 32 000 ha of forest. The fire started on August 15, 2023, due to a storm of scattering lightning strikes across the Six Rivers National Forest. Flames crossed Oregon, delaying the start of the school year for about 4000 students. Dry weather and gusty winds favored the flame propagation; dead lives and other dried vegetation of the forest floor functioned as tinder [67]. In Canada, among the several fire events starting in spring, burning more than 13.7 million ha, those affecting the Yellowknife, in the Northwest Territories, caused the evacuation of more than 20 000 people in mid-August [68]. At the Hawaii Islands (USA), wildfires started on August 8, 2023, which were favored by the dry, hot, and windy conditions, killing at least 114 people at Maui. Flames devastated the historic resort city of Lahaina, burning 880 ha of vegetation, and forcing tens of thousands of residents and tourists to evacuate the island [69]. In Europe, an intense fire season took place in Greece, Spain, Portugal, and Italy. In Greece, after the destructive events of Rhodes and Corfu recorded in July, several wildfires occurred the following month. The strongest fire event started on 19 August in the northeast region, near the city of Alexandropoulos, lasting several days because of dry vegetation, strong winds, and high temperatures. According to EFFIS, which supports the services in charge of the protection of forests against fires in the European Union (EU), flames burned more than 94 000 ha of vegetation [70].

A huge wildfire also affected Tenerife (Canary Island) since August 15, 2023, lasting about ten days, burning thousands of acres of woodland and forcing the evacuation of thousands of people [71]. An intense bushfire season in Australia (Southern hemisphere) was then recorded from April to November 2023, with about 33.0 million ha burning in the Northern Territory. In this region, which received only 0.4 mm of rain in September of 2023, strong winds, and dry and warm weather caused a rapid spread of flames [72].

Fig. 1 displays the geographic location of the wildfires investigated in this work, while Table I details these features in terms of vegetation cover and total burned area.

IV. DATA

The OLI/OLI2 pushbroom sensors, aboard the L8/9 satellites, which offer a combined revisit time of eight days (U.S. Geological Survey (USGS), 2023), deliver data in nine spectral bands with a narrow swath width of 185 km. Band 5 (NIR, 0.8 μm), band 6 (SWIR1, 1.6 μm), and band 7 (SWIR2, 2.2 μm), having a spatial resolution of 30 m, were frequently used to map hot targets (e.g., [38], [39], [47], [48], [49], [50], [51]). The OLI2, which provides internal calibration sources to ensure radiometric accuracy (ACC) and stability, thanks to a higher radiometric resolution than OLI (14-bit versus 12-bit quantization), appears more sensitive to subtle differences over darker areas (e.g., water or dense forests) [73]. The MSI, aboard the Sentinel-2A/2B satellites (five-day revisit time at the equator, thanks to a swath width of 290 km), provides

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multispectral images in 13 different spectral bands. Band 8A (NIR), band 11 (SWIR1), and band 12 (SWIR) mimic the OLI bands, with an improved spatial resolution (20 m).

Although those sensors do not have the same instrumental features, the global average revisit interval of about 2.3 days from Landsat 8, Landsat 9, and S2 virtual constellation (e.g., [74]), may be exploited to better investigate active fire areas with SWIR-based detections (e.g., [15]).

To perform this work, we processed the L8/9 Level 1-Standard Terrain Corrected (L1TP) products, including files with the scene-based metadata and the quality assessment band, made available by USGS and delivered the Earth Explore platform (https://earthexplorer.usgs.gov/). Moreover, we processed the S2 Level 1-C (L1C) products, including tiles 100 × 100 km^2 ortho-images in UTM/WGS84 projection, provided by the Copernicus Browser (https://browser.dataspace.copernicus.eu/). The Sentinel Application Platform (SNAP) and Quantum Geographic Information System (QGIS) software packages were used to generate the fire maps. Moreover, we analyzed the MSI and OLI/OLI2 data under GEE to perform the extended time series analysis over the Brady Hot Spring geothermal area reported in the Appendix.

V. Methods

The NHI algorithm uses two NHI to map high-temperature features [39]

$$\mathrm{NHI}_{\mathrm{SWIR}} = \frac{L_{\mathrm{SWIR2}} - L_{\mathrm{SWIR1}}}{L_{\mathrm{SWIR2}} + L_{\mathrm{SWIR1}}} \tag{1}$$

$$\mathrm{NHI}_{\mathrm{SWNIR}} = \frac{L_{\mathrm{SWIR1}} - L_{\mathrm{NIR}}}{L_{\mathrm{SWIR1}} + L_{\mathrm{NIR}}} \tag{2}$$

where L_{SWIR1}, L_{SWIR2}, and L_{NIR} are the TOA radiances measured in the SWIR1 (1.6 μm), SWIR2 (2.2 μm), and NIR (0.8 μm) bands of OLI and MSI sensors.

The $\mathrm{NHI}_{\mathrm{SWIR}}$ index, such as another normalized index independently proposed for hyperspectral data [75], exploits the major increment of the spectral radiance from hot targets such as lava flows and fires at the longer than shorter SWIR wavelength (e.g., [76]). Then, positive values of the $\mathrm{NHI}_{\mathrm{SWIR}}$ index generally characterize these features unlike background and cloudy areas. On the other hand, in the presence of more intense thermal anomalies, which frequently saturate the SWIR2 band of MSI and OLI, negative values of the $\mathrm{NHI}_{\mathrm{SWIR}}$ index may be recorded [39], [62]. By combining this index with the $\mathrm{NHI}_{\mathrm{SWNIR}}$, which assumes positive values over those features, the NHI algorithm performs an effective mapping of high-temperature targets (e.g., [39], [62], [77], [78]).

A. NHI Algorithm Tailored to Fire Mapping

Although the NHI algorithm may be used also to map wildfires, it could be less effective in the presence of less intense fire events, leading to negative values of the $\mathrm{NHI}_{\mathrm{SWIR}}$ index.

Indeed, the radiance measured by a sensor in daytime conditions, in the presence of a fire within the field of view (FOV), may be written as follows:

$$L_\lambda = \tau_\lambda \left(L_{\lambda b} + L_{\lambda f} + L_{\lambda r}\right) + L_{\lambda p} \tag{3}$$

where $L_{\lambda b}$ is the background-emitted radiance, $L_{\lambda f}$ the radiance emitted by the fire, $L_{\lambda r}$ the reflected solar radiance at the surface, $L_{\lambda p}$ the path radiance (which is a function of the atmospheric scattering of the reflected and emitted radiance and atmospheric emission), and τ_λ the atmospheric transmittance [79]. The radiance emitted by a surface at the temperature T is equal to

$$L_\lambda = B_\lambda(T) = \varepsilon_\lambda \frac{2hc^2}{\lambda^5 \left(e^{\frac{hc}{k\lambda T}} - 1\right)} \tag{4}$$

where λ is the wavelength, ε_λ the spectral emissivity, c(2.99792458 × 10^8 m/s) the speed of light, h (6.62607004 × 10^{-34} m^2 kg/s) the Planck's constant, and k (1.38064852 × 10^{-23} m^2 kg/s^2 K) the Boltzmann's constant. In the presence of a sub-pixel fire, at the temperature T_f, which occupies a fraction f of the pixel area, with a background component $(1 - f)$ at the temperature T_b, the emitted radiance is equal to

$$L_{\lambda E} = L_{\lambda f} + L_{\lambda b} = f B_\lambda\left(T_f\right) + (1 - f) B_\lambda(T_b). \tag{5}$$

The reflected solar radiance at the surface from the unburned component depends on the spectral reflectance ρ_λ and the solar irradiance E_λ according to the relation

$$L_{\lambda r} = \frac{(1 - f)\rho_\lambda E_\lambda \cos\theta}{\pi d^2} \tag{6}$$

where d is the Earth-sun distance and θ is the sun zenith angle. By considering the low contribution of the background emitted radiance at the SWIR wavelengths, and neglecting the path radiance and the reflected component from the burning fraction, we can approximate the radiance at the sensor in the SWIR band as follows:

$$L_\lambda = \tau_\lambda\left(L_{\lambda r} + L_{\lambda f}\right). \tag{7}$$

Therefore, the $\mathrm{NHI}_{\mathrm{SWIR}}$ index may be written as follows:

$$\begin{aligned} &\mathrm{NHI}_{\mathrm{SWIR}} \\ &= \frac{L_{\lambda_2} - L_{\lambda_1}}{L_{\lambda_2} + L_{\lambda_1}} \\ &= \frac{\tau_{\lambda_2}\left(L_{\lambda_{2r}} + f B_{\lambda_2}\left(T_f\right)\right) - \tau_{\lambda_1}\left(L_{\lambda_{1r}} + f B_{\lambda_1}\left(T_f\right)\right)}{\tau_{\lambda_2}\left(L_{\lambda_{2r}} + f B_{\lambda_2}\left(T_f\right)\right) + \tau_{\lambda_1}\left(L_{\lambda_{1r}} + f B_{\lambda_1}\left(T_f\right)\right)} \end{aligned} \tag{8}$$

where $L_{\lambda 2}$ and $L_{\lambda 2}$ (W m^{-2} sr^{-1} μm^{-1}) are the SWIR2 and SWIR1 radiances measured by satellite. Since the atmospheric transmittance exceeds 90% in the SWIR region (only upward atmospheric transmission is considered here for the emissive component), with a little dependence on atmospheric water vapor variations (e.g., [80], [81]), reflected solar radiance, fire fraction and temperature of hot targets appear as the main factors affecting the $\mathrm{NHI}_{\mathrm{SWIR}}$ index.

Fig. 2 displays the results of a simulation analysis, performed starting from (8), by assuming the fire and atmospheric parameters detailed in Table II. The plot shows the rapid increment of the $\mathrm{NHI}_{\mathrm{SWIR}}$ index up to positive values for $T_f \geq$ 1000 K and very small fire fractions, in the presence of a background composed only of oak vegetation (e.g., $\mathrm{NHI}_{\mathrm{SWIR}} \approx$ 0.05 for T_f = 1200 K and f = 0.4%; i.e., 3.6 and 1.6 m^2 for OLI and MSI, respectively).

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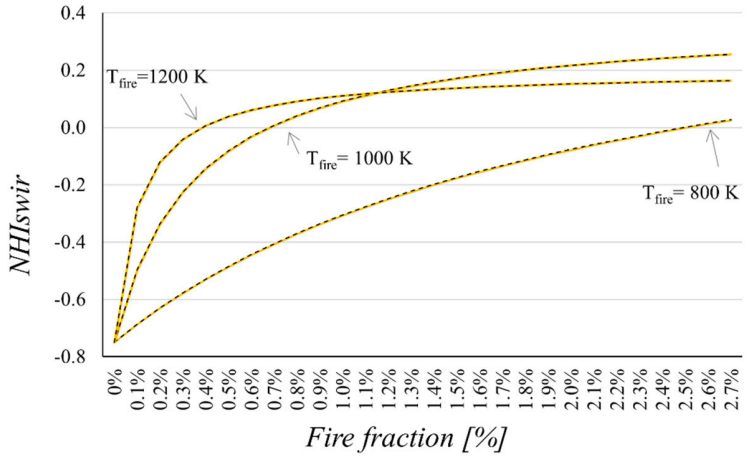


Fig. 2. Curves of the NHI_{SWIR} index retrieved, starting from (3)–(8) for different fire fractions and temperatures, by assuming the same parameters detailed in Table II (the spectral response function of OLI and MSI sensors is not considered here). Note the rapid increment of the index from negative to positive values for $T_f = 1200$ K.

TABLE II

INPUT PARAMETERS FOR THE SIMULATION ANALYSIS SHOWN IN FIG. 2 BY CONSIDERING THE REFLECTANCE VALUES [82], FLAME EMISSIVITY [83] ATMOSPHERIC TRANSMITTANCE [84] AND SUN IRRADIANCE VALUES IN THE SWIR BANDS FOR MSI AND OLI [42] DETAILED IN THE SECOND COLUMN

Parameter	*Value*
Sun zenith angle (θ) [rad]	0
Mean Sun-Earth distance (AU)	1
Transmittance τ_{SWIR1}	0.9
Transmittance τ_{SWIR2}	0.85
Flame emissivity $\varepsilon_{f\,SWIR1} = \varepsilon_{f\,SWIR2}$	0.9
Vegetation (oak) reflectance ρ_{SWIR1}	0.16
Vegetation (oak) reflectance ρ_{SWIR2}	0.07
OLI Sun Irradiance λ_{SWIR1} [W/m^{2}μm]	247.6
OLI Sun Irradiance λ_{SWIR2} [W/m^2 μm]	85.46
MSI Sun irradiance λ_{SWIR1} [W/m^2 μm]	245.6
MSI Sun Irradiance λ_{SWIR2} [W/m^2 μm]	85.25

TABLE III

VALUES OF THE NHI_{SWIR} RETRIEVED FROM (10), BY ASSUMING THE SAME PARAMETERS OF TABLE II, FOR TYPICAL BACKGROUND COVERS

Background	ρ_{SWIR1}	ρ_{SWIR2}	*OLI*	*MSI*
Vegetation (oak)	0.16	0.07	−0.75	−0.75
Vegetation (gray pine)	0.15	0.07	−0.74	−0.73
Vegetation (aspen)	0.30	0.17	−0.69	−0.69
Sand	0.16	0.12	−0.61	−0.61
Burned area	0.15	0.21	−0.37	−0.37
Soil	0.37	0.29	−0.59	−0.59
Melting Snow	0.02	0.02	−0.51	−0.51
Ice	0.11	0.16	−0.36	−0.35
Grass	0.28	0.19	−0.64	−0.64
Sea water	0.02	0.02	−0.51	−0.51

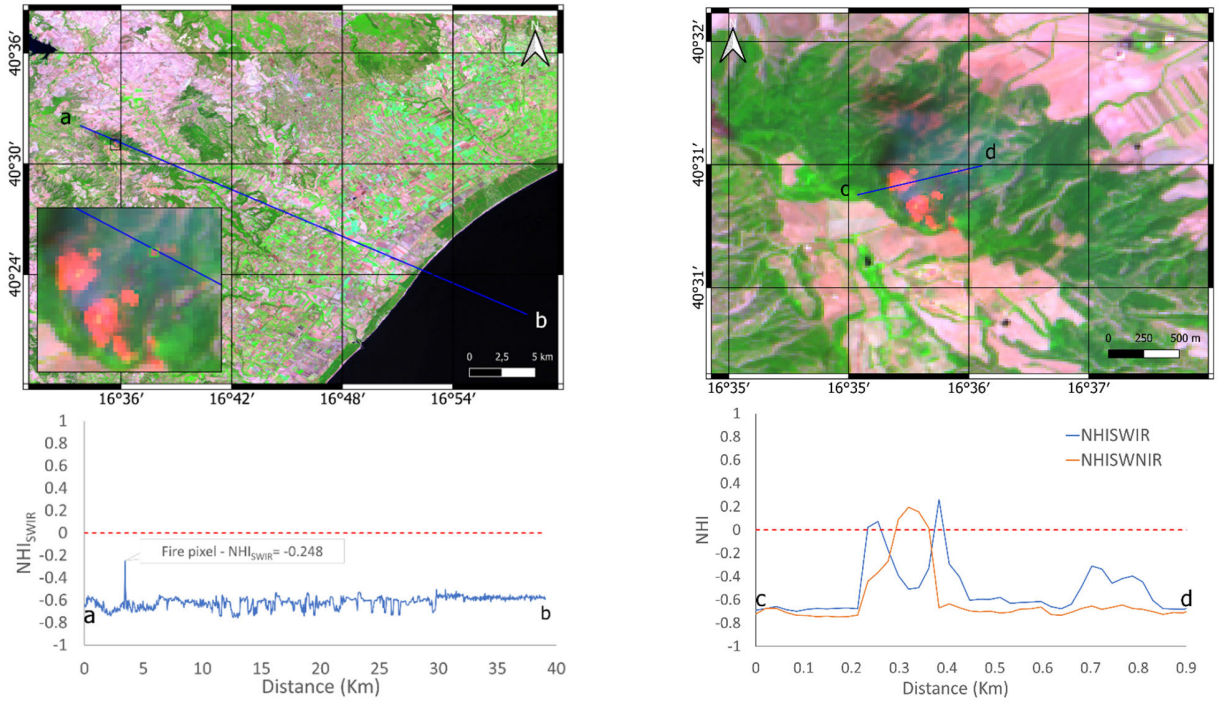
On the contrary, negative values of the index seem to characterize flaming fires of the same size and lower temperature affecting the same background (e.g., $NHI_{SWIR} = -0.53$ and $T_f = 800$ K).

The plot also shows a progressive increment of the index for $T_f = 1000$ K and $f > 1\%$ in comparison with the curve of $T_f = 1200$ K. Hence, a negative value of the NHI_{SWIR} index other than characterizing the background (e.g., $NHI_{SWIR} = -0.74$ in the absence of the fire component) because of higher radiance values measured at SWIR1 than SWIR2 wavelengths, may occur when the reflected solar component is still dominant (e.g., in presence of very small fire-fraction areas) or the thermal source is particularly radiant at the SWIR1 wavelengths (e.g., gas flaring sources with temperatures above 1600 K). This behavior may be then exploited to improve the identification of small fires if a proper detection threshold is used.

Table III details the NHI_{SWIR} values for a typical background cover, retrieved from the reflectance spectrums provided by USGS (Spectral Library Version 7; [82]) after setting the same parameters of Table II. The range $-0.3 < NHI_{SWIR} < -0.8$ appears as representative of the background conditions (see Table III). Negative values of the NHI_{SWIR} different from the background should then characterize flaming fires of small size. This behavior, which is derived from a simplified model (e.g., spectral response function of used sensors and saturation radiances are not considered here), was corroborated by the performed experimental analyses. An example is shown in Fig. 3, displaying the false RGB (red = B12; green = B11; blue = B8A) product from S2-MSI data of July 23, 2022, at 09:35 UTC, covering part of the Basilicata (Southern Italy) region.

The transect, extending from land to sea areas (*a*-*b* transect), intersects a small fire (<0.2 ha) affecting the vegetated area located close to the Pomarico Village (Matera; Southern Italy) [85]. The plot shown in the figure's bottom displays the spatial

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(a) (b)

Fig. 3. (a) S2-MSI false RGB (red = B12; green = B8A; blue = B4) subset image of July 23, 2022 at 09:35 UTC showing (top panel) small fire (red pixels) occurring close to the Pomarico (Matera, Southern Italy) village, and (bottom panel) spatial profile of the NHI_{SWIR} index retrieved along the *a*–*b* transect in blue. Note that the small fire pixel was undetectable by the standard NHI algorithm because of negative values of the NHI_{SWIR} (see red-dotted line). (b) Spatial profiles of the NHI_{SWIR} (blue curve) and NHI_{SWNIR} indices (orange curve) retrieved along the *c*–*d* transect marked in blue on top panel.

profile of the NHI_{SWIR} retrieved along the same transect, revealing a background level mostly in the range $-0.8 < NHI_{SWIR} < -0.5$, and the increment of the index above values of -0.3 just in correspondence of a fire pixel associated with the above-mentioned event (see false color image displayed in background). It should be stressed that the same wildfire was undetectable by the standard NHI algorithm because of negative values of the NHI_{SWIR} [see red dotted line in Fig. 3(a)]. Over the areas where the flaming fire was more intense [$c-d$ transect in Fig. 3(b)], the NHI_{SWIR} increased up to positive values, while it dropped under values of -0.4 in the presence of SWIR2 band saturation (see blue curve). In this circumstance, the NHI_{SWNIR} showed an opposite behavior, complementing information from the NHI_{SWIR} [see orange curve in Fig. 3(b)].

Pixels with values of this index higher than -0.3 affected about 0.001% of the entire satellite scene, as indicated by the analysis of Fig. 4, and were less affected by industrial hotspots, building rooftops, and spurious effects than pixels associated with lower critical values of the same index (e.g., $NHI_{SWIR} > -0.4$).

Thus, an enhanced NHI algorithm configuration exploiting values of $NHI_{SWIR} > -0.3$ could be more suited to discriminating small flaming fires from other targets in daylight conditions, regardless of background cover.

In night-time (i.e., in the absence of the reflected solar component), both NHI should assume positive values in the presence of active fires, due to differences in the emitted radiance at NIR and SWIR wavelengths. However, as for the daytime data, a spectral test on SWIR2 radiance should be used to filter out possible spurious effects [39].

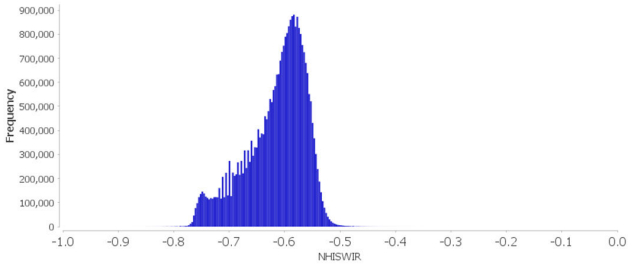


Fig. 4. Histogram of the NHI_{SWIR} pixels retrieved from the entire S2-MSI scene of Fig. 2(a), covering part of Basilicata, Puglia, and Calabria regions (Southern Italy). Note that 99.99% of the image pixels were in the range $-0.8 < NHI_{SWIR} < -0.5$.

In detail, by taking into consideration the results of some recent studies (e.g., [48], [86]), a threshold lower than that used in [38] should enable effective filtering of noisy pixels in night-time conditions, without affecting the identification of small fires, when combined to the NHI.

Hence, the NHI-F uses the following tests to map active fires through S2-MSI and L8/9 OLI/OLI2 data.

Daytime (S2 and L8/9 data)

$$\left\{\left(L_{\lambda_{2.2}} > 2.0\frac{W}{\mathrm{m^2 sr}\ \mu\mathrm{m}}\right)\ \text{AND}\ (NHI_{SWIR} > -0.3)\right\}: \text{HI}$$

$$\left\{\left(L_{\lambda_{2.2}} > 2.0\frac{W}{\mathrm{m^2 sr}\ \mu\mathrm{m}}\right)\ \text{AND}\ (NHI_{SWNIR} > 0)\right\}: \text{MI}$$

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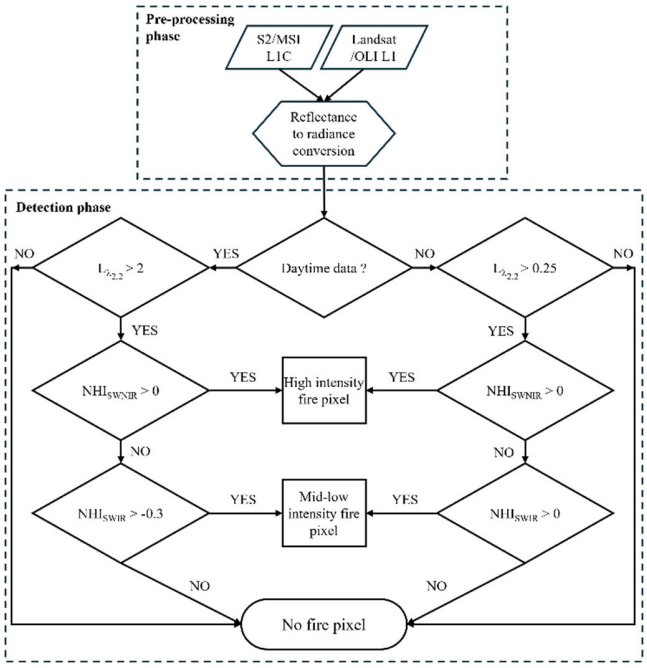


Fig. 5. Flowchart of the NHI-F algorithm configuration.

Night-time (L8/9 data)

$$\left\{\left(L_{\lambda_{2.2}} > 0.25\frac{W}{\text{m}^2\text{sr}\ \mu\text{m}}\right)\ \text{AND}\ (\text{NHI}_{\text{SWIR}} > -0)\right\}: \text{HI}$$

$$\left\{\left(L_{\lambda_{2.2}} > 0.25\frac{W}{\text{m}^2\text{sr}\ \mu\text{m}}\right)\ \text{AND}\ (\text{NHI}_{\text{SWNIR}} > 0)\right\}: \text{MI}$$

where HI and MI stand for high and mid-low intensity fire pixels, respectively. Fig. 5 displays the flowchart of the NHI-F configuration including both pre-processing (i.e., conversion from TOA reflectance to radiance) and detection phase. The latter is performed by implementing a threshold on SWIR2 radiance lower than that used to investigate the Kilauea's (Hawaii, USA) lava lake through night-time OLI data [87].

B. Assessment of Fire Detections

To assess the NHI-F detections, we used information from different sources (e.g., Copernicus Emergency Management Service (EMS); Northern Australia Fire Information) and the false color imagery emphasizing hot targets through the combination of the infrared bands (e.g., red = SWIR2, green = SWIR1; blue = NIR). The manual inspection of satellite imagery (e.g., [52]) was then performed using also other spectral bands (e.g., VIS, panchromatic) to assess the presence of fire smoke. In addition, we used the very high-resolution images available in Google Earth to verify the presence of some targets (e.g., solar panels, building rooftops, industrial sites) as a possible cause of false detections.

Based on these investigations, we used the following metrics to quantify the NHI-F performance:

$$P = \frac{t_p}{t_p + f_p} \tag{9}$$

$$R = \frac{t_p}{t_p + f_n} \tag{10}$$

$$\text{FDR} = \frac{f_p}{t_p + f_P} \tag{11}$$

$$\text{FNR} = \frac{f_n}{t_p + f_n} \tag{12}$$

where P is the precision, i.e., positive predictive value (PPV), R is the recall, true positive rate (TPR) or sensitivity, FDR is the false discovery rate ($1 - P$), and FNR is the false negative rate ($1 - R$). These metrics were retrieved from true positives t_p(fire pixels correctly detected), false positives f_p (nonfire pixels erroneously classified as fires), and false negatives f_n (actual fire pixels classified as nonfire). Two of these metrics (i.e., P and R) were used also by other authors to assess the performance of some machine learning models in the identification of volcanic hotspots for comparison with the NHI algorithm [88]. Here, we did not consider the ACC, since true negatives t_n (i.e., the correctly undetected nonfire pixels) were highly prevalent over the satellite scenes. As an example, the ACC was always above 99%, appearing less indicative of the algorithm performance (e.g., [56]). Obviously, since the ground truth at the pixel level was not available, metrics were affected by the used validation products.

VI. Results

In this section, we show the results of the fire mapping achieved by implementing the NHI-F on S2-MSI and L8/9 OLI/OLI2 data. Maps display the fire pixels in two different colors, based on fire intensity level (i.e., red = mid–low; yellow = high), the SWIR2 image in the background, and the fire perimeters from independent sources used as a reference to assess the fire mapping. We also performed a comparison with other systems, including those using satellite data at high temporal resolution, to better evaluate the added value of NHI-F detections.

A. Mapping of Active Fires Through S2-MSI Data

Fig. 6(a) shows the NHI-F map generated from S2-MSI scene of August 23, 2023, at 09:06 UTC (12:06 LT), covering the forestry region located northwest of Athens (Greece), where an intense wildfire occurred (Table I). The figure also displays the pixels (in red) flagged far from the fire perimeter (in green) provided by the Copernicus EMS.

The wildfire occurring in Greece affected the foothills of Mount Parnitha, destroying about 6193 ha of vegetation (mostly forest and shrubs), and forcing the evacuation of the population. The NHI-F algorithm flagged the fire pixels within the fire perimeter and in the surroundings of the burned area, as shown in Fig. 6(b), which was independently estimated around 3027 ha [89]. Pixels within the fire perimeter were consistent with a thin fire front emphasized by the false color image of Fig. 6(c), extending for about 16.8 km [89]. The Copernicus EMS reported other active flames also inside and in the neighbor of the main fire perimeter, corroborating most of the sparsely distributed fire pixels flagged by the NHI-F (see red pixels within the cyan circles), which identified some fire pixels also close to the fire perimeter. These pixels for the spatial continuity with the main front, reasonably represented true positives. Based on false negatives, including

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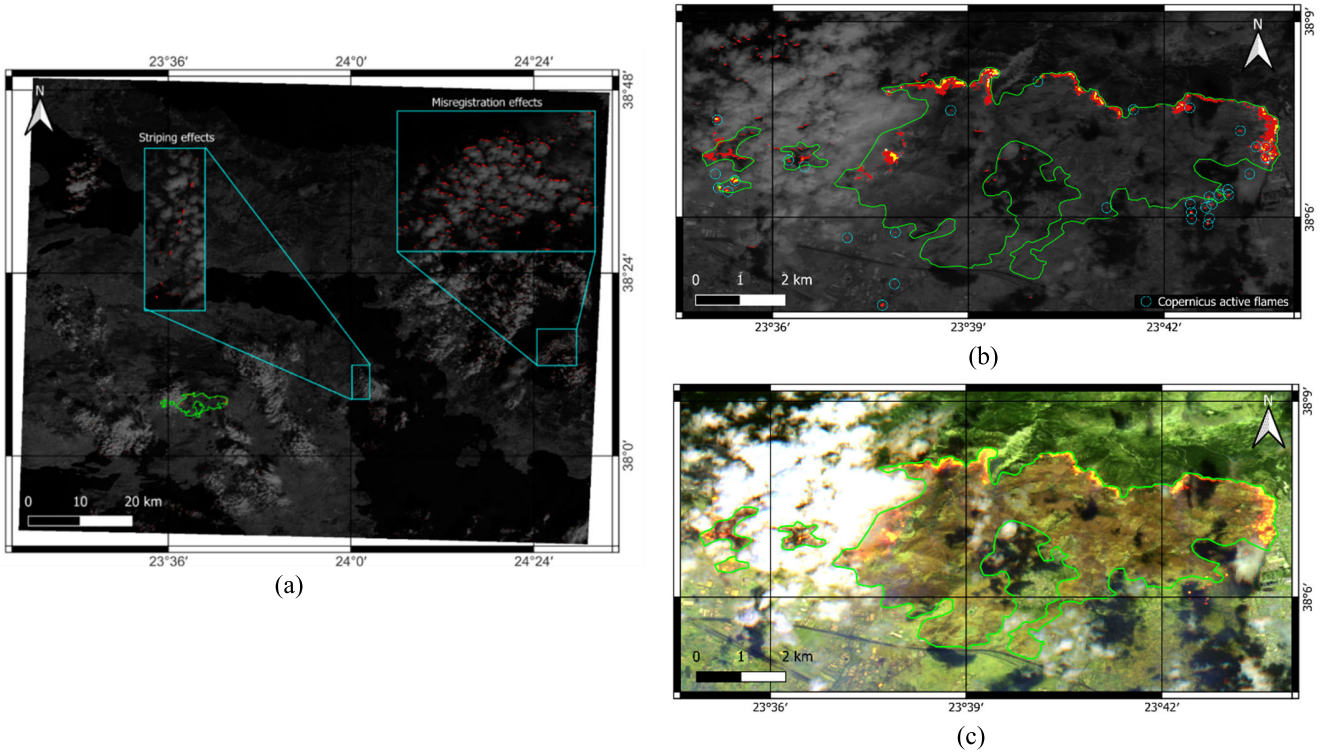


Fig. 6. (a) NHI-F map from S2-MSI scene (5490 × 5490 pixel in size) of August 23, 2023 at 09:06 UTC showing fire pixels (in red/yellow) associated with the intense wildfire affecting Greece, delimited by the perimeter in green from the Copernicus EMS [89], and regions with evidence of multispectral misregistration and striping (see red pixels within the areas marked in cyan). (b) Zoom of detected wildfire, with indication of other areas (in cyan), located also outside the fire perimeter, affected by active flames according to EMS. (c) False RGB (red = B12; green = B11; blue = B4) image.

pixels showing NHI_{SWIR} values slightly lower than the used threshold and pixels associated with the SWIR band saturation, we estimated a high sensitivity level ($R = 96.2\%$), and a low FNR (FNR = 3.8%). Concerning the pixels flagged far from the main front, they mostly affected cloud-edges or were aligned along the SW–NE direction. These pixels represented false positives associated with the multispectral misregistration and striping of S2-MSI imagery (e.g., [90]). Because of those data issues, the P was low ($P = 10.1\%$) and FDR particularly high (FDR = 89.9%).

To minimize their impact on NHI-F detections, which is assessed in more detail in the Appendix, we first used the spectral procedure implemented within the NHI tool [62]. This procedure, on the one hand, strongly reduced false positives (e.g., FDR = 1.5%) and, on the other hand, determined a significant increment of false negatives (FNR = 42.5%). Hence, we verified the possibility of minimizing the impact of multispectral misregistration and striping directly from the NHI. Fig. 7 displays, on the left panel, the scatter plot between the NHI_{SWIR} and NHI_{SWNIR} indices of the pixels flagged hot in Fig. 6(a) (i.e., with $NHI_{SWIR} > -0.3$). Most of these pixels showed values of $-1 < NHI_{SWNIR} < -0.7$ (see orange dots) and corresponded to the areas clearly affected by multispectral misregistration and striping marked in orange on the right panel. By excluding among the pixels those showing values of $NHI_{SWNIR} < -0.7$, we observed a significant reduction of false positives (FDR = 6.8%) with a slight increment of false negatives (FNR = 6.5%), mostly corresponding to the active flames located outside the fire perimeter, as shown in Fig. 8. Despite the occurrence of minor striping effects, this approach could be then used to minimize the impact of S2-MSI data issues on NHI-F detections, with a negligible impact on fire mapping ACC, regardless of monitored areas and thanks to the stable behavior of NHI (e.g., [91]).

On the other hand, the performance of the NHI-F algorithm strongly increased when multispectral misregistration and striping effects were less significant.

This is evident in Fig. 9, displaying the NHI-F map from S2-MSI scene of August 18, 2023, at 11:52 UTC. In more detail, Fig. 9(a) shows the fire pixels associated with the Arafo Candelaria fire, affecting the Tenerife Island (see fire perimeter from EMS in green), and a few false detections magnified on the top-right panel (A–D panels). Fig. 9(b) focuses on the active fire region emphasized by the false color image (left panel) used to assess the NHI-F detections (right panel). The suspected arson, which was declared over three months later, destroyed about 15 000 ha of forest (pine woodland covers about 60% of the forestry area of the Canary Islands [92]), blanketing the island with smoke and ash [71], [93]. The NHI-F detections were consistent with the information from the false color image, showing that dense fire smoke, dispersing in the E-NE direction, also affected the investigated area. This feature unlike smearing effects (i.e., adjacency effects from highly radiant pixels on nearby nonfire areas, e.g., [87]), causing a slight overestimation of the fire front, did not impact NHI-F performance. In general, minor striping effects,

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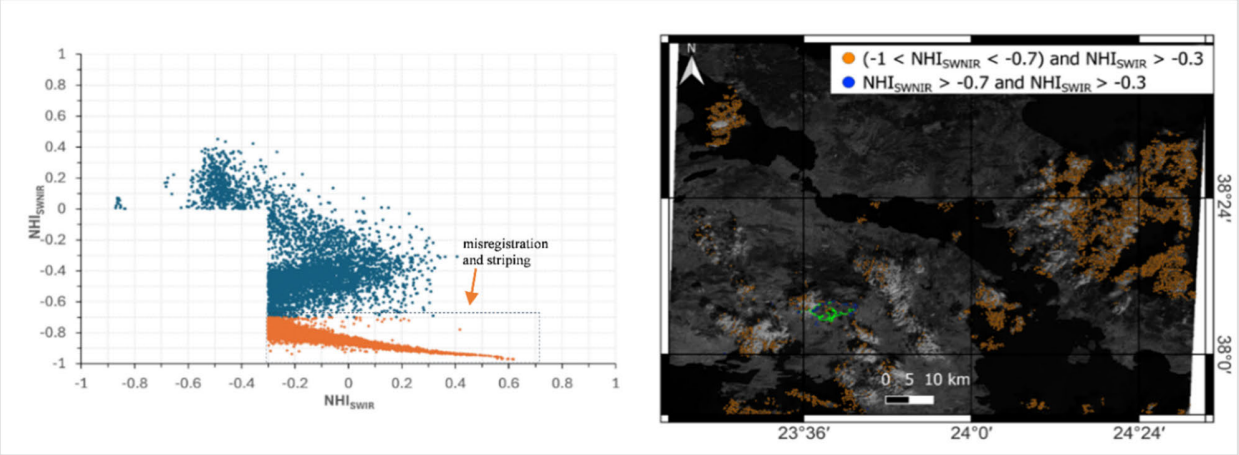


Fig. 7. (Left panel) Scatter plot between the NHI_{SWIR} and NHI_{SWNIR} indices over the pixels flagged hot by (right panel) NHI-F on S2-MSI scene of August 23, 2023 at 09:06 UTC. Orange dots indicate pixels with values of $NHI_{SWIR} > -0.3$ and $-1 < NHI_{SWNIR} < -0.7$ mostly corresponding to multispectral misregistration and striping marked in orange on the right panel (in background the SWIR2 image); blue dots indicate pixels ascribable to the active flames.

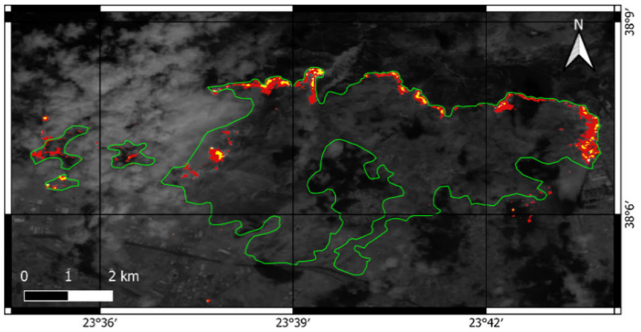


Fig. 8. NHI-F map from S2-MSI scene of August 23, 2023 at 09:06 UTC focusing on the active fire region of North Attica (Greece), generated by filtering out pixels with values of $NHI_{SWNIR} < -0.7$ (see text); in background the SWIR2 image.

solar panels (panel A), and building rooftops (panels B–D) represented the main cause of false detections [see Fig. 9(a)], while the single hot pixels flagged over the Atlantic Sea, close to the coasts of Tenerife Island (panel E), was ascribable to a fast-moving object (see also Appendix). Based on the results of these analyses, we found that NHI-F showed high performance (e.g., $P = 97.2\%$, $R = 98.5\%$) in the mapping of Arafo Candelaria fire in daylight conditions.

It is worth mentioning that by exploiting the multispectral misregistration of S2-MSI imagery (e.g., [94], [95]), i.e., measuring the displacement of the object in the panel E and considering the temporal offset (1 s) of B02 and B04 bands, we estimated a velocity of about 360 Km/h, which is compatible with a Canadair [96], which was probably extinguishing flames over the Tenerife Island at the time of S2 overpass.

To assess the NHI-F performance in climatic and vegetation conditions different from Southern Europe, we investigated some recent wildfires occurring in Australia (see Table I).

In the Northern Territory, landscapes are dominated by the tropical savannah, consisting of sparsely treed grassland, open forest, and eucalypt woodland, which are subjected to widespread fire activity because of a quick desiccation during the dry season [97]. Fig. 10 shows the results retrieved from S2-MSI data of July 13, 2023, at 01:37 UTC (LT-9:30), covering the Darwin region. The map includes the fire perimeters provided by North Australia and Rangeland Fire Information (NAFI), which delivers hotspot products from National Oceanic and Atmospheric Administration (NOAA), National Aeronautics and Space Administration (NASA), and European Space Agency (ESA) satellites. Information about burned areas in the Northern Territory is sourced from the Darwin Centre for Bushfires Research. Most fire pixels flagged hot by the NHI-F (i.e., ∼97%) affected the vegetated areas delimited by the fire perimeter or were in spatial contiguity with the main flaming fronts, as indicated by the A−B panels marked in cyan and magnified on the right side of the figure. The remaining pixels affected other vegetated areas (panels C) or were flagged over industrial sites and building rooftops (panels E–H) (see very high-resolution imagery displayed in the background). A misdetection associated with a fast-moving object also occurred (panel D). Due to a higher number of false negatives, ascribable to weak active flames, we observed a slight reduction of the algorithm performance in comparison with the Arafo Candelaria fire (e.g., $R = 93.2\%$; $P = 96.8\%$). It is worth mentioning that a persistence analysis, devoted to characterizing the gas flaring sites on a global scale, showed that some nude soils (e.g., those possibly containing iron oxides) may lead to positive values of the NHI_{SWNIR} index [64]. This aspect should be considered when the NHI-F is used to map wildfires over other regions of Australia (e.g., Western Australia).

Table IV summarizes the metrics retrieved by analyzing the above-mentioned fire events. These metrics quantify the NHI-F performance in the identification and mapping of active fires through S2-MSI data, showing for instance the P in the range of 93%–97% (after using the filter for multispectral misregistration; see Fig. 7) and the R in the range 93%–99%.

B. Active Fire Mapping Through L8/9 Data and Comparison With the Operational LFTA Product

To evaluate the efficiency of NHI-F detections when L8/9 OLI/OLI2 data are used, we performed a comparison with the

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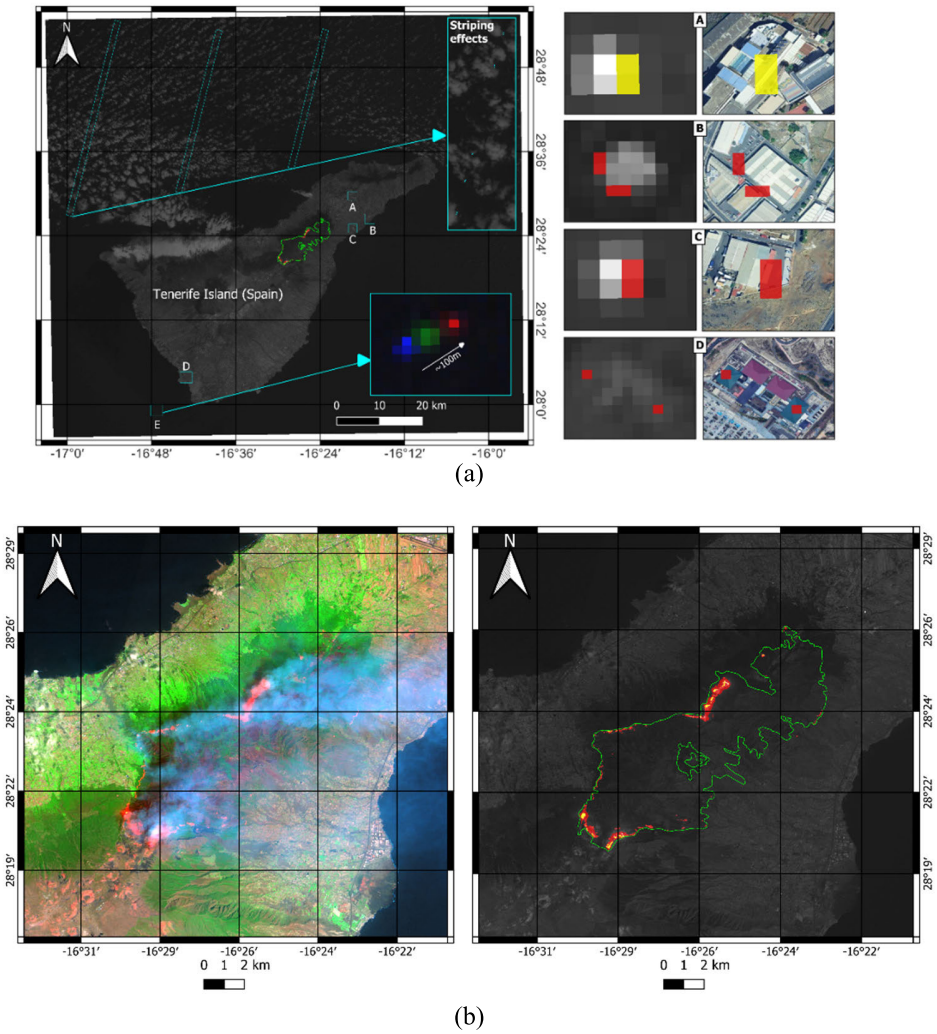


Fig. 9. (a) NHI-F map from S2-MSI scene of 18 August, 2023 at 11:52 UTC (LT + 2) covering the wildfire affecting the Tenerife Island (Spain). In background, the SWIR2 image, the panel E (red = B12; green = B11; blue = B4 image in background) shows a fast-moving object (see text). On the right, false positives associated with building rooftops and solar panels (A–D). (b, Left panel) False color RGB (red = B12; green = 8A; blue = B4) image and (right panel) NHI-F detections yellow/red pixels). In green, the fire perimeter from Copernicus EMS.

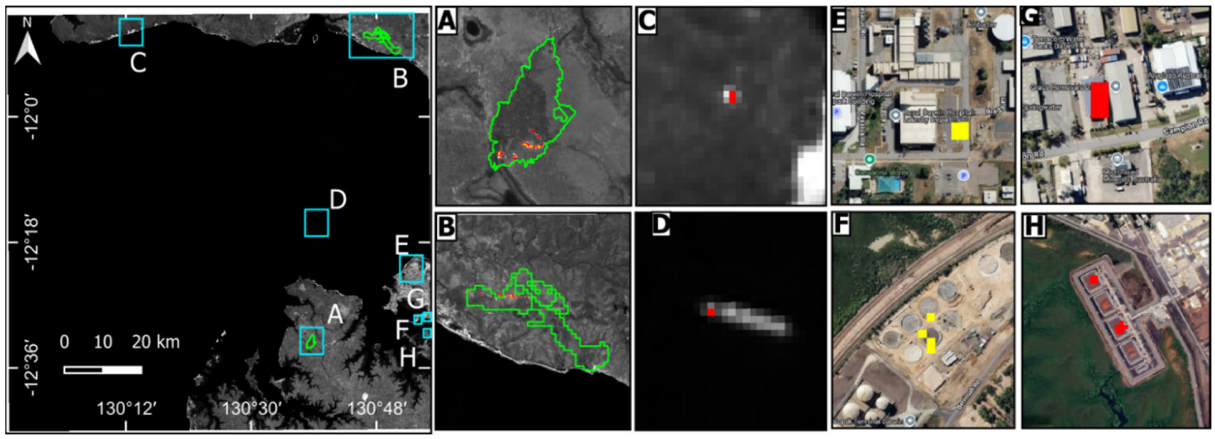


Fig. 10. On the left, NHI-F map from S2-MSI data of July 13, 2023 at 01:37 UTC covering the Darwing Region (Australia), in background the B12 (2.2 μm) image, with the zoom of the active fires delimited by NAFI (green perimeter) (panels A and B) and other features (panels C–H) described in the text.

operational LFTA product. The USGS and U.S. Department of Agriculture Forest Service (SDA) generate this fire and thermal anomaly product based on a method previously developed for ASTER [37], and the algorithm proposed by [38] for all the daytime L8/9 acquisitions. The availability of nighttime data is generally much lower; moreover, those data are

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TABLE IV
METRICS RETRIEVED BY ANALYZING ACTIVE FIRES THROUGH THE NHI-F CONFIGURATION

S2-MSI data	*Fire Location*	*P*	*R*	*FDR*	*FNR*
23 August 2023 09:06 UTC	North Attica (Greece)	10.1%	96.2%	89.9%	3.8%
23 August 2023 09:06 UTC	North Attica (Greece) with filter	93.2%	93.5%	6.8%	6.5%
18 August 2023 11:52 UTC	Tenerife (Spain)	97.2%	98.5%	2.8%	1.5%
13 July 2023 01:37 UTC	Northern Territory (Australia)	97.0%	93.2%	3.0%	6.8%

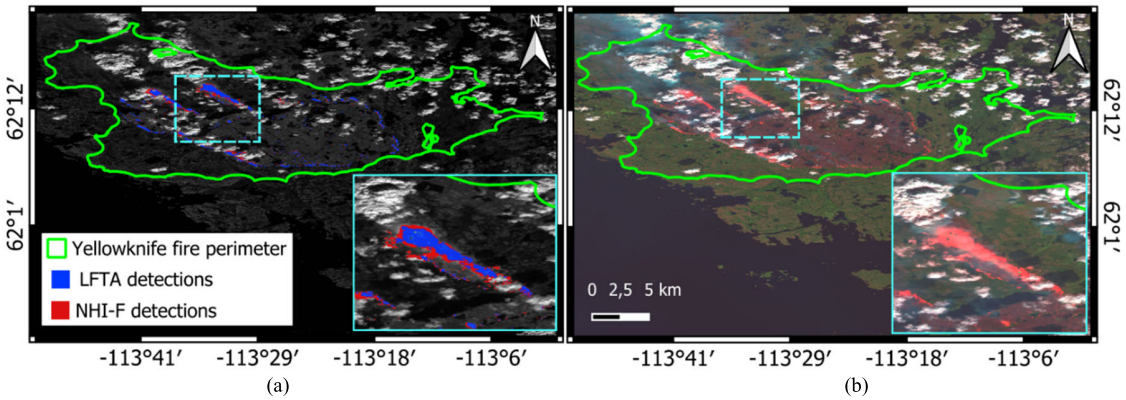


Fig. 11. Wildfire affecting the Yellowknife (Canada) identified on daytime L8-OLI scene of August 9, 2023 at 18:45 UTC. (a) LFTA (in blue) and NHI-F (in red) detections. In the inset of the figure, zoom of the northern part of the fire front.(b) False color image (red = B7; green = B5; blue = B4). In green, the fire perimeter from CWFIS database.

more frequently acquired over specific targets (e.g., active volcanoes). The FIRMS database, which distributes the NRT active fire data from MODIS and VIIRS at a global scale within 3 h of satellite observation, delivers the LFTA product over Continental U.S. (CONUS), Canada, and northern Mexico. In this study, we focused our analyses over Yellowknife and California, using daytime data from FIRMS, and implementing the spectral test detailed in [38] in night-time conditions.

Many studies have shown that the fire regimes in the boreal forest are increasing in terms of extent, severity, and duration (e.g., [98]). The large wildfire affecting the Yellowknife investigated in this work burned about 50 750 ha of vegetation during the period 2 August 24−September 2023 (https://cwfis.cfs.nrcan.gc.ca/home). Information from the Canadian Wildland Fire Information System (CWIFS) database is used here as a reference. The CWIFS monitors the fire danger conditions and fire occurrence across Canada, generating the fire perimeters from satellite data at mid-low spatial resolution (e.g., MODIS and VIIRS). Fig. 11 shows the LFTA and NHI-F detections [Fig. 11(a)] in different colors, performed on daytime L8-OLI data of August 9, 2023, at 18:45 UTC (12:45 LT) and the relative false color subset image [Fig. 11(b)]. About 98.7% of LFTA detections (i.e., 8175 fire pixels) were coincident with those from the NHI-F. The latter missed about 1.3% of LFTA fire pixels mainly because of SWIR band saturation effects. Nonetheless, the unique NHI-F detections were significantly higher in number (8537 pixels). They better detailed the fire front (e.g., see the area marked in cyan magnified in the figure inset), as indicated by the manual inspection of satellite data and the information provided by the false color image of Fig. 11(b). This comparison shows the capacity of NHI-F in improving the mapping of active fires affecting the boreal forest.

To assess differences in terms of fire mapping with the LFTA product over California, we accessed the information from the Wildland Fire Interagency Geospatial Services (WFIGS). This database provides geospatial data products under the interagency Wildland Fire Data Program, hosted in the National Interagency Fire Center ArcGIS Online Organization (NIFC Org) (https://data-nifc.opendata.arcgis.com/), by detailing the fire parameters, such as the fire discovery date and time, the primary and secondary fuel and the fire size.

Fig. 12(a) displays the fire maps generated from the L9-OLI2 scene of August 24, 2023, at 18:57 UTC (LT+7), displaying the LFTA (blue bordered pixels) and NHI-F (red pixels) detections. All the fire pixels were correctly detected within the WFIGS perimeters (in green), which delimited the Kelly, Holiday, Coon, and Hurdy Gurdy fires. The Kelly fire, starting in the afternoon of August 15, 2023, merged with the Holiday fire and affected a large portion of the Smith River Complex, burning about 32 900 ha of vegetation. The wildfire caused economic losses estimated around U.S. $146 955 000 [99], [100].

The NHI-F algorithm detected about 99% of the LFTA fire pixels (i.e., 1216 over 1230). Regarding the exclusive NHI-F detections (1413 fire pixels), they were mostly in spatial continuity with the main fire fronts, as indicated by the region magnified on the bottom-left panel (see areas 1 and 2). To validate the fire pixels flagged within the WFIGS perimeter at a higher distance from the main fronts, we performed some additional investigations. Specifically, we assessed the presence of fire smoke and computed the delta normalized burn ratio (dNBR) index to evaluate the fire severity

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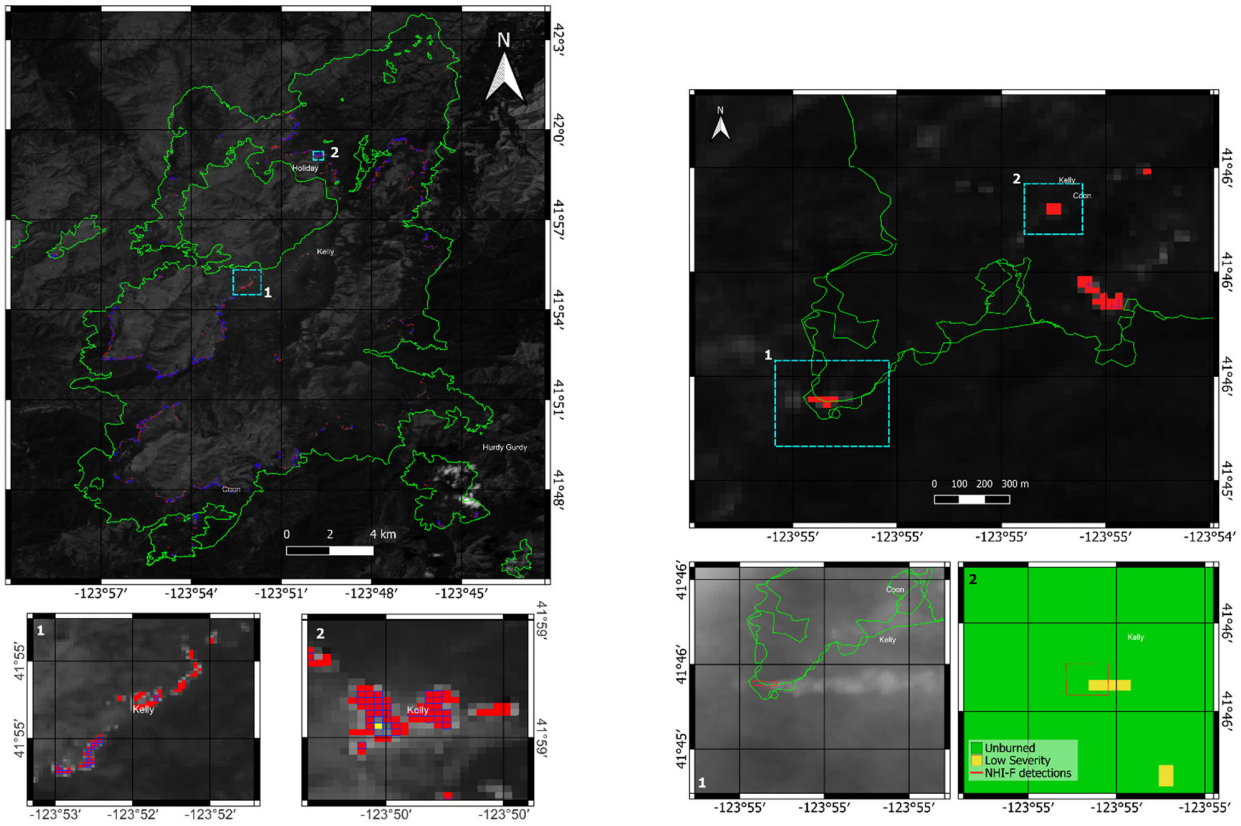


Fig. 12. (a) LFTA (blue bordered pixels) and NHI-F (red pixels) detections performed on daytime L9-OLI2 scene of 24 August at 18:57 UTC, in green the fire perimeter from WFIGS, in background the SWIR2 image. On the bottom, areas marked by the dotted rectangles with evidence of some unique NHI-F detections. (b) Exclusive NHI-F fire detections performed over the areas marked in cyan, located at larger distance from the main front. On the bottom, magnification of the area-1 (with the panchromatic band displayed in background) showing a fire smoke plume dispersing in the W–E direction, and area-2 displaying the dNBR map from S2-MSI data of 15 and 25 August, 2023 (low severity in yellow; unaffected areas in different shades of green).

(e.g., [101], [102], [103], [104]). In Fig. 12(b), we show an example of this analysis performed over two distinct regions of the Kelly fire (see regions marked in cyan on the top panel). By analyzing the OLI2 panchromatic band (15 m spatial resolution), we found that a smoke plume, dispersing in the SE direction, affected area-1 (see bottom-left panel), and indirectly corroborated the presence of active flames over that area. Moreover, the analysis of S2-MSI scenes of 15 and 25 August at 18:59 UTC, acquired before and soon after the Landsat-9 overpass of Fig. 12(a), shows that a low fire severity characterized the area-2 (see yellow region), just in correspondence of detected fire pixels (see red bordered pixels), corroborating the NHI-F detections at a higher distance from the main front.

Differences with the operational LFTA product were even more significant in night-time conditions, as shown in Fig. 13. The figure displays the NHI-F [Fig. 13(a)] and LFTA [Fig. 13(b)] products from L9-OLI2 scene of August 26, 2023, at 05:57 UTC (22:57 LT). It is worth noting that the NHI-F identified most pixels appearing bright, i.e., those which were more radiant than neighbor ones in the SWIR2 band shown in the background (e.g., see right panels), ascribable to the active flames, well characterizing them also in terms of relative intensity level (see red/yellow pixels). Moreover, the NHI-F detections (4202 fire pixels) were spatially contiguous with the LFTA ones (1805 fire pixels) or affected other vegetated areas still located within the WFIGS perimeter, which were reasonably associated with small fire outbreaks. Table V quantifies the results of these analyses, showing the increment of exclusive NHI-F detections up to about 70% in night-time conditions, when very small flaming fires should be better detectable in the SWIR bands [38]. The important contribution offered by the night-time OLI/OLI2 observations to the fire mapping was even more evident for comparison with TIR Sensor (TIRS) data of the same day and hour, which did not enable an effective discrimination of small fires from the background.

Despite these advantages, nighttime OLI/OLI2 data are scarcely available [38], suggesting their usage mainly as a complement to daytime observations, as shown and discussed in detail in Section VI-C.

C. Fire Mapping and Characterization Through S2 and L8/9 Data Integration

The contribution of the NHI-F to the wildfire mapping may be particularly significant when S2 and L8/9 observations are integrated. In this way, the temporal coverage of the active fire areas may be increased, especially over areas

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TABLE V

COMPARISON BETWEEN NHI-F AND LFTA PRODUCT PERFORMED IN BOTH DAYLIGHT AND NIGHT-TIME CONDITIONS OVER THE YELLOWKNIFE (CANADA) AND CALIFORNIA (USA) BY ANALYZING PIXELS FLAGGED HOT WITHIN THE WFIGS PERIMETER. THE DIFFERENT PERCENTAGES WERE CALCULATED BY THE NUMBER OF TOTAL FIRE PIXELS FROM THE TWO PRODUCTS (E.G., 16 712 PIXELS FOR THE YELLOWKNIFE'S WILDFIRE), I.E., BY CONSIDERING BOTH COMMON AND EXCLUSIVE FIRE PIXELS

Fire Location	*Satellite overpass*	*Common fire pixels*	*Only NHI-F*	*Only LFTA*
Yellowknife	09/08/2023 18:45 UTC	48.3%	51.1%	0.6%
Kelly	24/08/2023 18:57 UTC	46.0%	53.5%	0.5%
Kelly	26/08/2023 05:57 UTC	30.0%	70.0%	0.0%

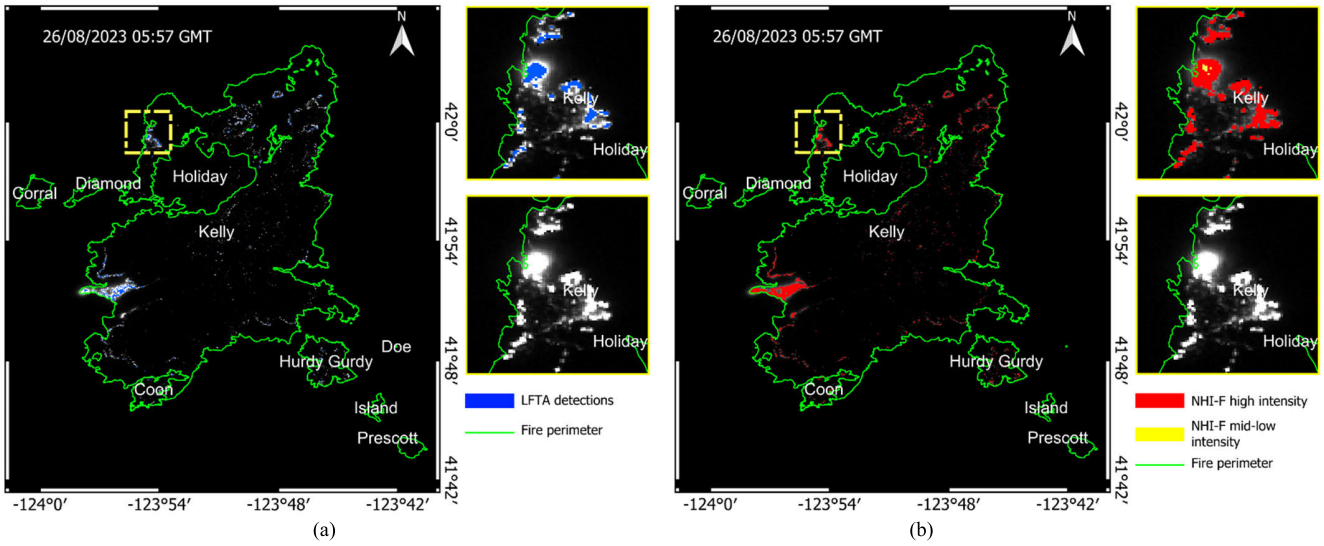


Fig. 13. Fire maps generated from the L9 OLI-2 data of August 26, 2023 at 05:57 UTC covering California analyzed through (a) spectral test detailed in [38] (LFTA, blue pixels) and (b) NHI-F algorithm (red pixels). In background the SWIR2 image. Bright pixels within the fire perimeter in green (from WFIGS) reveal the areas affected by active flames. On the bottom, magnification of the region of Kelly/Holiday fire marked by the dotted yellow square (top panels) with detected fire pixels and (bottom panels) without using any algorithm.

where higher is the availability of night-time OLI/OLI2 scenes from ascending L8/9 overpasses. An example is shown in Fig. 14, displaying the space-time evolution of the Kelly fire retrieved from consecutive S2 and L8/9 scenes of 15–August 30, 2023, covering California. Map highlights the progressive displacement of the fire fronts (see pixels depicted in different colors), within the WFIGS perimeter marked in black. In detail, since 15 August, when the first fire outbreaks were identified by satellite (see red pixels), the flaming front has spread in different directions, approaching the NE and W-SW border of the fire perimeter at the end of the same month (see magenta pixels). Therefore, even if the Kelly fire was contained in early November 2023, most vegetation reasonably burned during the initial phase of the fire event. The top-right panel displays the region of fire front within the yellow box, showing the prevalent average wind direction retrieved through data interpolation, during the period 18–26 August 2023, starting from information provided by the NASA–Prediction of Worldwide Energy Resources (NASA–POWER) (https://power.larc.nasa.gov/data-access-viewer/). Over this region of the Kelly fire the fire front displaced about 1500−1600 m, from point *a* (800 m elevation) to point *b* (450 m elevation) (see the elevation profile in the bottom-right panel) in about seven days, burning temperate needleleaf forest and shrubland, with a rate of fire spread estimated around 9.5 m/h. This value gives an indication of the speed of the fire front over the region marked in yellow in Fig. 14. Although more accurate models have been proposed to estimate the fire rate of spread by satellite (e.g., [105]), which is affected by several factors such as wind, fuel, weather and topography (e.g., [106]), this analysis shows that the integrated S2 and L8/9 observations may provide a suited contribution in this direction.

Fig. 15 displays the cumulated curve of the active fire area, retrieved from the number of NHI-F fire pixels, by assuming the whole pixel area is entirely affected by flames. The major increment (∼600 ha) of this parameter was recorded in between 25 and 26 August, when the Kelly fire appeared spatially more extended. Hence, more than 1600 ha of vegetation burned during the first 15 days of the monitored fire event.

However, the plot provides a partial overview of the phenomenon under investigation due to several factors. They include: 1) the limited temporal sampling of used data (i.e., just one image per day at maximum); 2) the presence of meteorological clouds partially or completely obscuring the hot targets; 3) a partial coverage of the investigated area at the edge of some L8/9 night-time scenes; and 4) the occurrence

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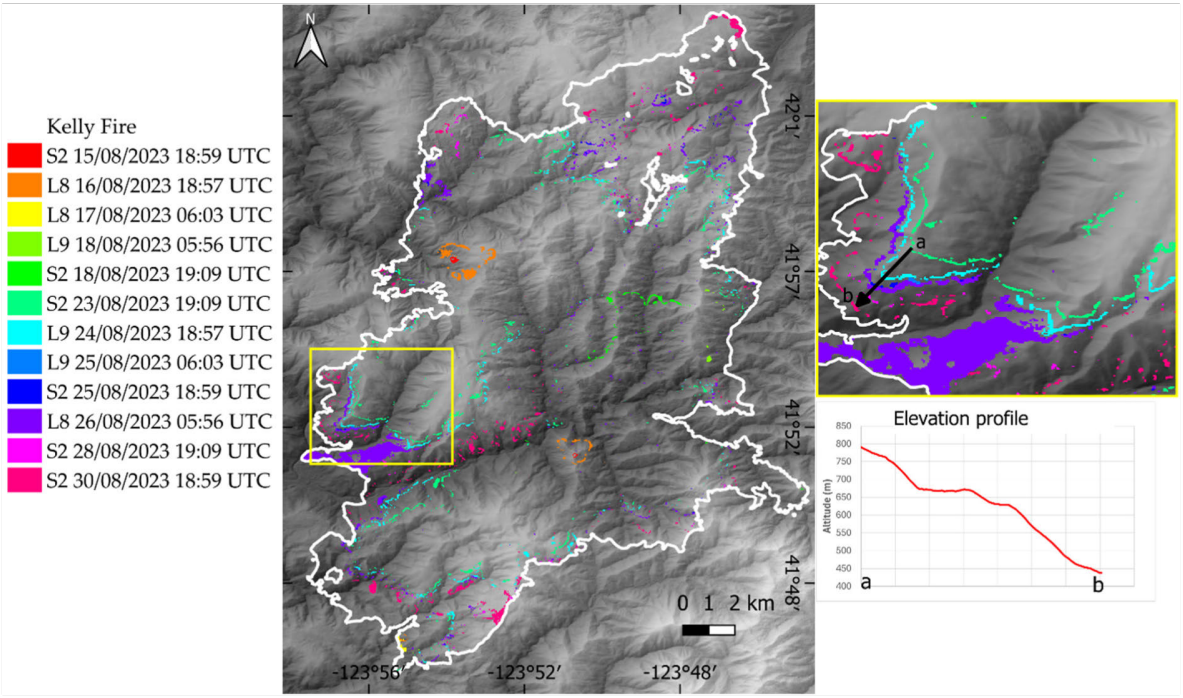


Fig. 14. Map of the Kelly fire (California; USA) generated by overlapping the fire pixels flagged by the NHI-F algorithm on daytime (S2 and L8/9) and night-time (L8/9) satellite scenes of 15−30 August, 2023. The different pixel colors indicate a different satellite overpass date and time. In background, the Copernicus 30 m global digital elevation model (DEM) analyzed under QGIS, using the hill-shade function. On the right, magnification of the area in the yellow box showing the displacement of the fire front of about 1500–1600 m from 18 to 26 August, 2023, and the mean wind direction marked by the black arrow. The bottom right panel displays the elevation profile derived from the DEM, showing the displacement of the fire front from about 800 to 450 m elevation.

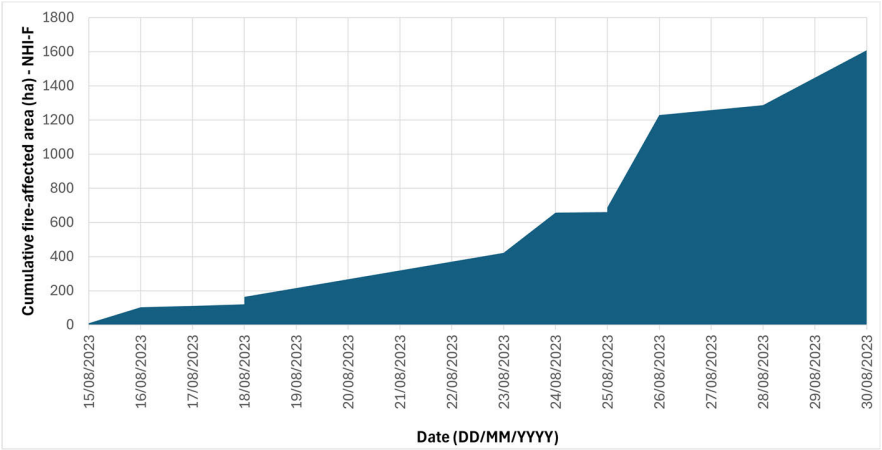


Fig. 15. Cumulated curve of the area covered by the active flames investigating the Kelly of Fig. 14 through daytime (S2 and L8/9) and night-time (L8/9) satellite data of 15–30 August, 2023 by means of the NHI-F algorithm.

of pixel saturation effects. Indeed, despite the advantages of the combined S2 and L8/9 observations, wildfires require the analysis of satellite data at a high temporal resolution to be efficiently monitored (e.g., see period 18−23 August, 2023). Moreover, while cloud coverage represented the major factor affecting the identification of wildfires, the SWIR band saturation had a minor impact on the results of this work.

The NHI-F products, generated from S2 and L8/9 data integration, should be then used mainly as support to operational fire systems, performing the NRT monitoring of wildfires (see next section).

VII. Discussion

Results of the comparison with the LFTA product performed over Yellowknife and California show that NHI-F may improve the mapping of active fires in both daylight and night-time conditions. This improvement may be exploited to complement information from fire monitoring systems using satellite data at high temporal/low spatial resolution.

Fig. 16 shows an example of this potential, in reference to the Olinda fire (Hawaii Islands), which started on August 8, 2023, at around 12:30 A.M. (10:30 UTC) burning about 438 ha of vegetation (mainly tall grass and brush). The fire was ignited by a downed power line hit by a tree, because of

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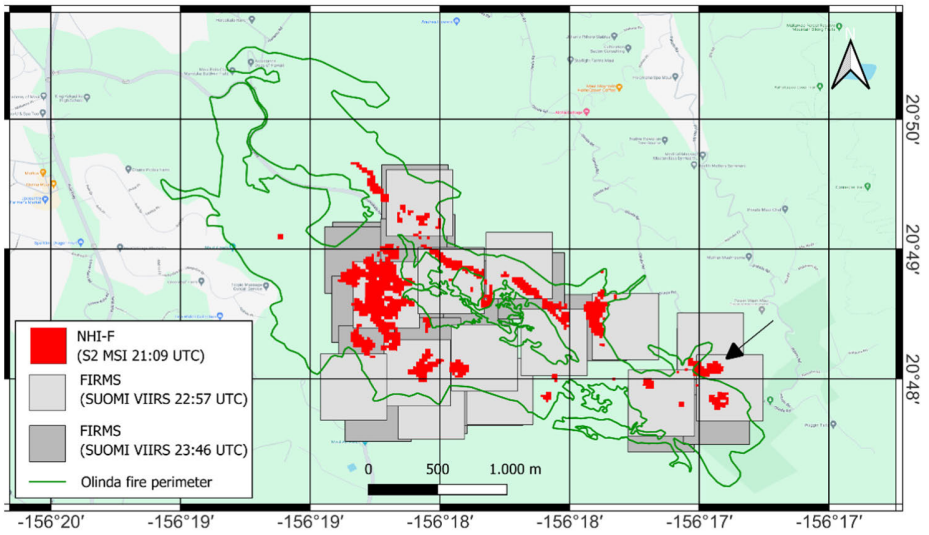


Fig. 16. NHI-F and NASA-FIRMS detections, from daytime VIIRS data of August 8, 2023 at 22:57 UTC (in light gray) and 23:46 UTC (in dark gray), overlapped on the map of Olinda fire. In green, the wildfire perimeter from WFIGS. The black arrow indicates the fire pixels flagged slightly outside the fire perimeter, suggesting the occurrence of a slightly more extended fire over that area.

strong winds [107]. While the catastrophic Maui wildfire was investigated by other authors using multiresolution global fire products from MODIS, VIIRS, and PlanetScope [108], here we assess the added value of S2-MSI observations in providing accurate information about areas affected by the active flames, when analyzed through a proper fire detection method.

In detail, the figure displays the fire pixels (in red) flagged by the NHI-F, on S2-MSI scene of August 8, 2023, at 21:09 UTC (LT+10), and those detected by FIRMS, on VIIRS data of the same day at 22:57 UTC (light gray) and 23:46 UTC (dark gray), associated with the Olinda fire and delimited by the WFIGS fire perimeter. The NHI-F detections were mostly included within the fire perimeter and indicated that active flames extended over an area up to 45 ha at the time of satellite observation. The few fire pixels flagged outside the fire perimeter marked by the black arrow were consistent with the information from the false color image used to emphasize hot targets. Therefore, it is realistic to suppose that the Olinda fire was slightly more extended over that area. Moreover, the NHI-F provided unique information about some small fire outbreaks (e.g., see the isolated fire pixels detected both NW and SE of the main front), thanks to the higher spatial resolution of MSI than VIIRS data. On the other hand, FIRMS indicates that the active flames were approaching the southern region of the fire map a few hours after the S2 overpass (see gray pixels).

This analysis shows that the NHI-F may integrate information from systems performing the NRT monitoring of active fires by satellite, contributing to better delineating areas affected by active flames and detecting small fire outbreaks.

It should be stressed that NHI-F performance, quantified in Table IV, should be even higher when L8/9 OLI/OLI2 data are used. The analysis of those data, unlike the S2-MSI ones, did not show evidence of multispectral misregistration and/or striping effects (see also Appendix). Moreover, false positives were particularly low even in night-time conditions, as indicated by the metrics derived from the entire L9-OLI2 scene of August 26, 2023, at 05:57 UTC covering California (i.e., $P = 99.8\%$ and FDR $= 0.2\%$). In nighttime, false positives were mostly associated with industrial hotspots and fire accidents. Fig. 17 shows an example of these features in reference to the biogas flares (panels a and b), which are used to dispose of gases generated from organic matter decomposition (e.g., sewerage sludge), and a house fire observed by some surveillance video cameras (see panel c).

These results demonstrate the capacity of the NHI-F algorithm to perform well regardless of the used sensor, geographic area, vegetation cover, and observational condition.

A GEE-App implementing the NHI-F configuration is currently under development. The App, by exploiting the continuously updated dataset of satellite observations ingested in GEE [65], will enable the analysis of wildfires over the regions of interest with the benefits of the S2 and L8/9 data integration. Fig. 18 displays the beta version and some of the outputs generated by investigating the intense Maui wildfire of August 8, 2023. The procedure tested here to minimize multispectral misregistration and striping effects directly from the NHI, and the spectral tests used by the NHI tool to take into consideration possible pixel saturation effects, will be implemented within the App to increase the ACC of fire mapping. The NHI-F, which could also integrate the information from the daytime approach for gas flaring identification (DAFI) [64] system to better discriminate fires from gas flaring sources, could then provide a relevant contribution to the fire management activities also thanks to the rapid processing times offered by the GEE platform. Specifically, from a few seconds to a few minutes (over larger areas in the presence of more satellite imagery to process) are required to generate the NHI-F products under GEE.

From a scientific point of view, satellite data at mid-high spatial resolution are particularly suited to validate operational

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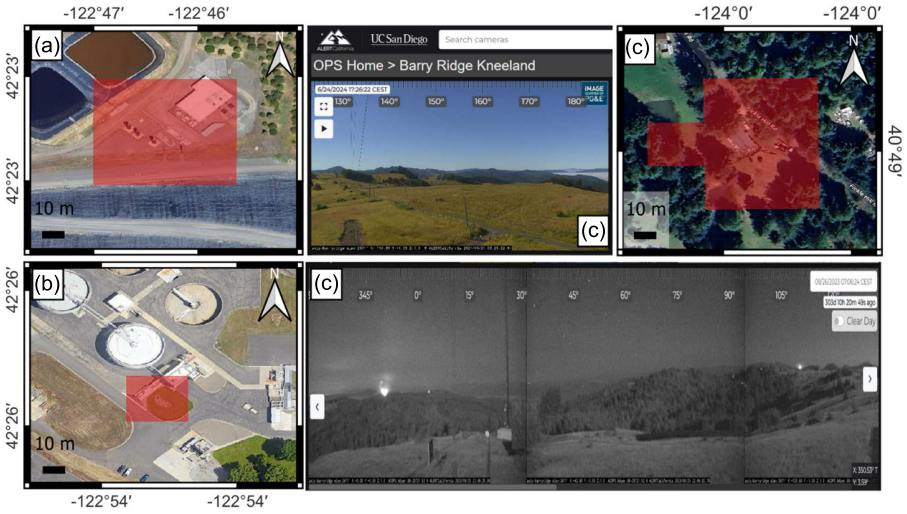


Fig. 17. Example of NHI-F detections outside the WFIGS perimeters in Fig. 13, associated with (panels a and b) waste flaring and (panel c) house fire revealed by alert surveillance camera (https://alertcalifornia.org/).

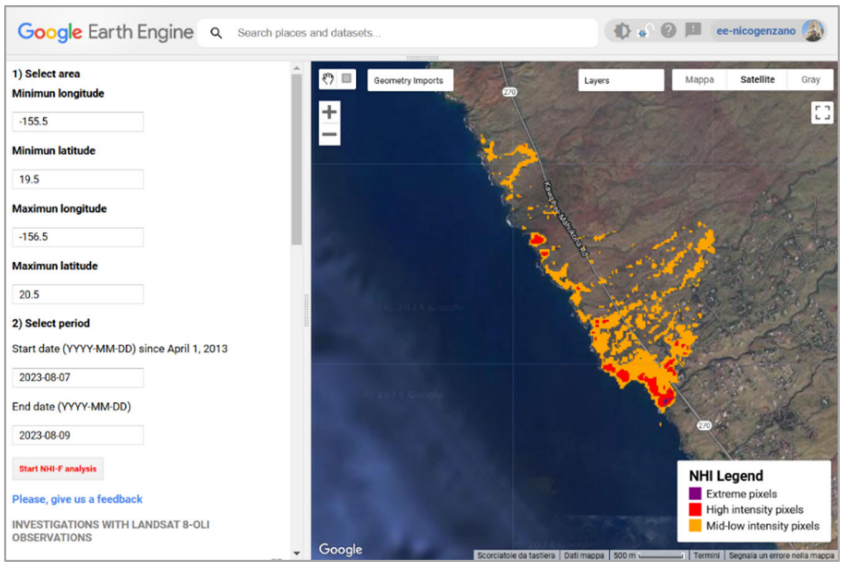


Fig. 18. Beta version of the GEE-tool integrating the NHI-F detection scheme, developed for the global fire mapping through S2 and L8/9 data. On the right, fire pixels, reported in three different colors based on relative intensity level, associated with the intense Maui fire (Hawaii, USA) mapped on S2-MSI scene of August 8, 2023 at 21:09 UTC.

fire detection products on a global scale, in the absence of field information (e.g., [26]). In this context, the NHI-F algorithm, thanks to a simple and effective detection scheme, which may be easily applied to NIR and SWIR data from different sensors, may highly support scientists in assessing the information retrieved from other sources also through its implementation in integrated satellite-based systems for fire detection and prioritization (e.g., [109]).

This study opens some interesting perspectives also regarding the possibility of using the NHI-F to monitor active fires in NRT through daytime and night-time SLSTR SWIR data. Those data could integrate information retrieved in the MIR band, at lower spatial resolution (1 km), as previously performed analyzing thermal volcanic features by means of GOES-R observations (e.g., [110]). A similar approach could be then used to investigate active fires, over Europe and Africa, with a very-high frequency of observation (10 min), through VIS and SWIR2 data, in the high-resolution fast imagery (HRFI) mode (500 m spatial resolution), from the Flexible Combined Imager (FCI), aboard Meteosat Third Generation (MTG) geostationary satellites [111].

VIII. Conclusion

In this work, we have presented and tested an enhanced configuration of the NHI algorithm tailored to fire mapping, by analyzing some harmful fire events occurring across the world during the intense fire season of 2023.

The NHI-F may be applied to L8/9 and S2 data without any optimization, enabling the identification of active fires in different background and observational conditions, as demonstrated by results retrieved over Greece, Tenerife, Australia, Canada, California, and Hawaii Islands.

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TABLE VI

S2-MSI SCENES OF DECEMBER 2018 – AUGUST 2023 SHOWING PIXELS WITH VALUES OF $NHI_{SWIR} > -0.3$. RESULTS WERE RETRIEVED BY EXCLUDING PIXELS WITH LOW SWIR2 RADIANCE VALUES (SEE SECTION II) AND USING THE S2QA CLOUD-MASKING PRODUCT TO FILTER OUT METEOROLOGICAL CLOUDS

Satellite Scene	*Pixel with values of $NHI_{SWIR} > -0.3$*	*Factors*
Jan 7, 2019	8	cloud-edges
Jun 21, 2019	6	cloud-edges/vehicle traffic
Jun 26, 2019	1	vehicle traffic
Jul 11, 2019	1	vehicle traffic
Jul 26, 2019	2	cloud-edges
Sept 29, 2019	7	cloud-edges
Feb 26, 2020	2	possible plant burning
Mar 2, 2020	2	possible plant burning
Mar 22, 2020	1	cloud-edges
May 31, 2020	1	vehicle traffic
May 31, 2021	4	vehicle traffic
Jun 10, 2021	2	vehicle traffic
Jun 30, 2021	74	cloud-edges
Jul 10, 2021	3	cloud-edges
Jul 25, 2021	44	cloud-edges
Jul 30, 2021	51	cloud-edges
Dec 27, 2021	6	cloud-edges
Apr 11, 2022	9	cloud-edges
Jul 10, 2022	3	cloud-edges
Jul 15, 2022	2	vehicle traffic
Sept 13, 2022	36	cloud-edges
Nov 2, 2022	6	cloud-edges
Dec 27, 2022	240	cloud-edges
May 1, 2023	16	cloud-edges
May 6, 2023	23	cloud-edges
May 21, 2023	2	cloud-edges
Jun 20, 2023	33	cloud-edges
Jun 25, 2023	1	cloud-edges
Jun 30, 2023	1	vehicle traffic
Jul 30, 2023	1	vehicle traffic

These results, assessed using information from independent sources/databases, demonstrate that the NHI-F may significantly improve the mapping of active fire areas in comparison with the earlier NHI algorithm, thanks to a better balance between reliability and sensitivity.

The main limitations of the NHI-F include the underestimation of active fires in the presence of thick cloud coverage, and the overestimation of fire fronts due to image artifacts (e.g., smearing effects). Indeed, multispectral misregistration and striping of S2-MSI imagery, representing the main factors causing false detections and affecting also other fire detection methods (e.g., [53]), although representing the main cause of false detections, may be minimized directly from the used indices, as demonstrated in this study. False positives from OLI/OLI2, showing minor striping in all the bands (e.g., [112]), were particularly low and mainly associated with solar panels, building rooftops (in the daytime), and industrial heat sources (night-time/daytime). These features are stable in space and can be easily identified and masked.

The comparison with the operational LFTA product, performed over Yellowknife and California, has shown an improved capacity of NHI-F in detailing the fire fronts and detecting small-fire outbreaks in both night-time and daylight conditions.

Finally, this study demonstrates the advantages of L8/9 and S2 data integration for a more continuous mapping of the active fire areas. To the best of our knowledge, this integration has never been performed so far by combining night-time and daytime imagery to characterize the fire dynamics and retrieve information about the rate of fire spread from satellite data at mid-high spatial resolution.

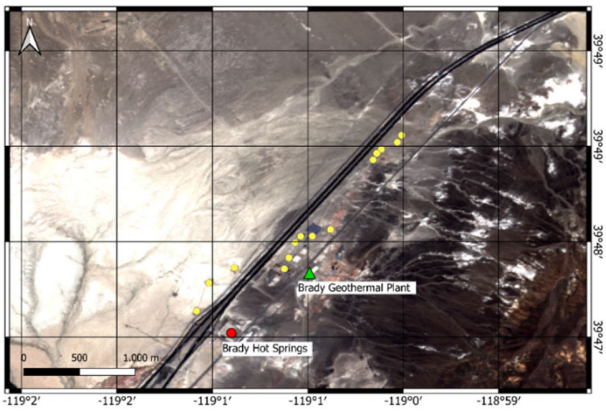


Fig. 19. Map of the Brady Hot Spring (red dot) area (Nevada, USA) with indication of the geothermal plant (green triangle). In yellow the production wells.

The possibility of using the normalized hotspot indices to detect fires by means of self-adaptive thresholds is also under evaluation, with the aim of further improving the identification and mapping of active fires from space.

APPENDIX

To better assess factors affecting the NHI_{SWIR} index, when negative values are explored to detect fires, we performed an extended confutation analysis over the Brady Hot Spring

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(a) (b)

Fig. 20. (a) Pixels showing values of $NHI_{SWIR} > -0.3$, on S2-MSI scene of June 21, 2019 at 18:43 UTC, associated with the vehicle traffic. (b) Thermal anomaly (red pixels) showing the same values of NHI_{SWIR} index, on S2-MSI scene of March 2, 2020 at 18:47 UTC, affecting the onion field of the food dehydration plant operating at the Brady Hot Spring geothermal area. In the inset, the false color image (red = SWIR2, green = SWIR1, blue = NIR), in background, the very high-resolution imagery (Maxar Technologies) available in GEE.

(Nevada, USA) geothermal area, which is located near the town of Fernley, in the northwestern Great Basin region (Fig. 19). The surface manifestations of geothermal activity at Brady Hot Spring area include sinter, warm ground, mud pots, and fumaroles [113], [114], [115], which for their temperature (<500 K) should be undetectable in the SWIR bands (e.g., [75]). A geothermal power plant of about 26 MW extracts geothermal fluids from six production wells (since 1992) to supply the onion drying facility (e.g., [116]). The plant uses geothermal water to wash down the processing equipment, and clean and dehydrate onions during the processing season (i.e., from May to December each year).

Table VI reports the list of satellite images showing pixels with values of $NHI_{SWIR} > -0.3$ over the Brady Hot Spring area, in reference to the period December 2018−August 2023. While L8/9 data did not reveal pixels with those values of the index, apart from a couple of satellite scenes presumably affected by calibration issues, pixels with values of $NHI_{SWIR} > -0.3$ affected about 8.8% of S2-MSI scenes (341 in total). By the manual inspection of satellite imagery, we found that none of the pixels was ascribable to geothermal features or possible fires. Indeed, pixels with the above-mentioned values of the index were mostly flagged over cloud-edges, unfiltered by the S2 Quality Assessment (QA) cloud-masking procedure used to mask opaque and cirrus clouds [117]. These pixels like those flagged over the Dwigth D. Eisenhower Highway were reasonably associated with the multispectral misregistration of S2-MSI data. The latter are acquired in each spectral band at a slightly different time, owing to the inter band parallax (angle and temporal sensing offset), related to the sensor builds geometry [118]. Consequently, the geometric shift recorded in the presence of high-altitude and/or fast-moving objects [95] may lead to false detections when SWIR bands are combined to detect fires (e.g., [62]). Fig. 20(a) displays for instance a false hot pixel associated with the vehicle traffic. A couple of satellite scenes in Table VI (26 February and 2 March, 2020) did not show, however, any evidence of multispectral misregistration or striping. By analyzing the very high-resolution imagery available in GEE, we found that these pixels were exactly located over the onion field of the food dehydration plant operating at the Brady Hot Spring geothermal area.

Since the S2-MSI imagery was acquired outside the processing season, when there is a reduced usage of geothermal water to dehydrate onion, we can speculate that pixels marked in red in Fig. 20(b), corroborated by the false color image shown in the figure inset, was associated with a possible burning of the onion plants. Hence, the NHI-F configuration, on the one side, is more affected than the standard NHI algorithm by the factors causing false detections (e.g., satellite data issues/artifacts) on the other side is more sensitive to vegetation fires, including those of a low-intensity level which are often associated with agricultural practices (e.g., [4], [118]).

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